# SAMS — Sprint 1 Test Cases & Test Data

*TASK-002 | Consolidated suite TC-001 to TC-045*

Project: Student Attendance Management System (SAMS)

Environment: Local FastAPI backend, static frontend, and SQLite test database.

## 1. Purpose and scope

This document consolidates 45 test cases and the test data used during local testing. Results are based on recorded observations. Smoke tests and cases needing deployment or mobile evidence are clearly qualified.

## 2. Test data

| Item | Test data | Notes |
|---|---|---|
| Admin | username admin | Local test account; password intentionally omitted. |
| Student | student1@example.com | Local test account; password intentionally omitted. |
| Faculty | faculty1@example.com | Local test account; password intentionally omitted. |
| Class | Test Class A, term 2026, ID 1 | Created successfully. |
| Student record | Test Student One, student1@example.com, ID 1 | Class ID 1. |
| Subject | TEST101 / Test Subject, ID 1 | Creation succeeded on retry after a transient SQLite lock. |
| Faculty record | Test Faculty One, faculty1@example.com, ID 1 | Created successfully. |
| Assignment | Faculty 1, subject 1, class 1; ID 1 | Created successfully. |
| Attendance session | 2026-10-26, period 1, session ID 1 | Created successfully. |
| Attendance | Student 1, session 1; Present then corrected to Absent | Duplicate mark returned HTTP 409. |
| Rules | Threshold 75%; correction window 48 hours | Returned by /api/rules. |
| Negative inputs | Invalid email, empty name, short password, Unknown status, unknown IDs, injection-style search | Used in validation/security checks. |

## 3. Test cases

PASS means expected behavior was observed. LIMITED SCOPE means only a smoke test was performed. PARTIALLY VERIFIED means more evidence is needed. PENDING means deployment/environment verification is outstanding.

| Test ID | Test case | Observed result | Status / notes |
|---|---|---|---|
| TC-001 | Admin valid login | HTTP 200; token and Admin role | PASS |
| TC-002 | Admin invalid password | HTTP 401 | PASS |
| TC-003 | Create class | HTTP 200; Test Class A, ID 1 | PASS |
| TC-004 | Create student | HTTP 200; Test Student One, ID 1 | PASS |
| TC-005 | List students | HTTP 200; active student returned | PASS |
| TC-006 | Invalid student email | HTTP 400 | PASS |
| TC-007 | Duplicate student email | HTTP 409 | PASS |
| TC-008 | Create subject | Initial HTTP 500 SQLite lock; retry HTTP 200, ID 1 | PASS WITH OBSERVATION |
| TC-009 | Create faculty | HTTP 200; faculty ID 1 | PASS |
| TC-010 | Create assignment | HTTP 200; assignment ID 1 | PASS |
| TC-011 | Create attendance session | HTTP 200; session ID 1 | PASS |
| TC-012 | Mark attendance Present | HTTP 200; one record saved | PASS |
| TC-013 | Duplicate attendance prevention | HTTP 409 Duplicate attendance | PASS |
| TC-014 | Attendance history | HTTP 200; history returned | PASS |
| TC-015 | Attendance percentage | HTTP 200; 100%, 1/1 before correction | PASS |
| TC-016 | Low-attendance list | HTTP 200; empty before correction | PASS |
| TC-017 | Attendance report | HTTP 200 | PASS |
| TC-018 | CSV export | HTTP 200 | PASS |
| TC-019 | Filter by Present status | HTTP 200 before correction | PASS |
| TC-020 | Correct Present to Absent | HTTP 200 | PASS |
| TC-021 | Audit log for correction | HTTP 200; correct_Present_to_Absent visible | PASS |
| TC-022 | Invalid attendance status | HTTP 400 | PASS |
| TC-023 | Unknown session ID | HTTP 404 Session not found | PASS |
| TC-024 | Student not enrolled in class | HTTP 400 | PASS |
| TC-025 | Attendance rules | HTTP 200; threshold 75%, correction window 48h | PASS |
| TC-026 | Date filter | HTTP 200 | PASS |
| TC-027 | Student access to admin audit | HTTP 403 | PASS |
| TC-028 | Logout invalidates token | Logout HTTP 200; reused token HTTP 401 | PASS |
| TC-029 | Unauthenticated student list | HTTP 401 | PASS |
| TC-030 | Faculty login | HTTP 200; role Faculty | PASS |
| TC-031 | Faculty attendance summary | HTTP 200 | PASS |
| TC-032 | Faculty attendance report | HTTP 200 | PASS |

| | | | |
|---|---|---|---|
| TC-033 | Response-time smoke test | /openapi.json HTTP 200 in 0.001702 seconds (single local request) | PASS — LIMITED SCOPE |
| TC-034 | Availability/recovery smoke test | HTTP 200 after backend restart | PASS — LIMITED SCOPE |
| TC-035 | Password storage | Salted PBKDF2-HMAC-SHA256, 200,000 iterations; salt$hash | PASS |
| TC-036 | Attendance data integrity | Duplicate mark returned HTTP 409 | PASS |
| TC-037 | UI login/dashboard | Valid login opened dashboard; 0%, 0 of 1 attended; history shows Absent for 2026-10-26, period 1, Test Subject | PARTIALLY VERIFIED — mobile not separately tested |
| TC-038 | Maintainability and automated tests | pytest: 2 passed, 18 warnings; main logic concentrated in backend/main.py | PASS WITH OBSERVATION |
| TC-039 | Concurrent requests smoke test | Five simultaneous /openapi.json requests all HTTP 200 | PASS — LIMITED SCOPE |
| TC-040 | HTTPS/TLS | HTTP 200; HTTPS request HTTP 000 on local dev server | PENDING DEPLOYMENT VERIFICATION |
| TC-041 | Password hashing and log redaction | Hashing verified; no print/logging/logger/basicConfig matches found in backend/main.py | PARTIALLY VERIFIED — log check limited to one file |
| TC-042 | Role-based authorization | Student token to admin-only /api/audit returned HTTP 403 | PASS |
| TC-043 | Input validation and injection | Invalid student data HTTP 400; injection-style search returned [] / HTTP 200; inspected search query uses placeholders and args | PASS FOR TESTED CHECKS |
| TC-044 | Protected audit logs | Student token to /api/audit returned HTTP 403 | PASS |
| TC-045 | Session timeout | With SAMS_SESSION_TIMEOUT=5, reused token after 7 seconds returned HTTP 401 Session expired | PASS |

## 4. Evidence and outstanding checks

- TC-037 screenshot: screenshot supplied by tester shows successful student dashboard login, 0% attendance, 0 of 1 classes attended, and history row 2026-10-26 / period 1 / Test Subject / Absent. Suggested evidence name: TC-037_Student_Dashboard.png.
- TC-037: Desktop login/dashboard verified. Mobile responsiveness has not been separately verified.
- TC-040: Local development server did not serve HTTPS. Verify TLS in the actual deployment environment before final acceptance.
- TC-041: Password hashing source inspected. Logging search was limited to backend/main.py; this is not proof of comprehensive log redaction.
- TC-033 and TC-039 are smoke checks, not formal performance or load tests.
- TC-038: pytest reported 2 passed and 18 warnings; review warnings separately.
- TC-008: Subject creation initially returned HTTP 500 during a SQLite lock, then succeeded on retry.

## 5. Submission notes

Do not include real passwords, bearer tokens, or secret values in submitted evidence. Confirm pending/partially verified cases with the team before formal submission.

## 6. Screenshot evidence index

The companion Test Results & Defects document includes the 40 evidence screenshots supplied earlier in this conversation (Evidence_01.jpg through Evidence_40.jpg), plus the latest student dashboard screenshot. They are retained as source evidence; individual screenshot-to-test mappings should be confirmed against the original execution sequence before submission.

Latest screenshot: Student dashboard after valid login. It shows 0% attendance, 0 of 1 classes attended, and an Absent history entry for 2026-10-26, period 1, Test Subject.

